Simplifying we get $$p^2 - q^2 = 2pq\left(\frac{1 - m_1m_2}{m_1 + m_2}\right)$$

$$\Rightarrow p^2 - q^2 = 2pq\left(\frac{1 - \frac{a}{b}}{-\frac{2h}{b}}\right)$$

That is $$\frac{p^2 - q^2}{a - b} = \frac{pq}{h}$$

$\therefore$ The locus of $P(p, q)$ is $$\frac{x^2 - y^2}{a - b} = \frac{xy}{h} \tag{6.36}$$

Example 6.36 Show that the straight lines $x^2 - 4xy + y^2 = 0$ and $x + y = 3$ form an equilateral triangle.

Solution:

Let the line $x + y = 3$ intersects the pair of line x^2 -4xy + $y^2 = 0$ at A and B. The angle between the lines $x^2 - 4xy + y^2 = 0$ is

$$\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a + b}\right| = \frac{2\sqrt{4 - 1}}{2} = \sqrt{3}$$
$$\Rightarrow \theta = \tan^{-1}\sqrt{3} = 60^\circ$$

The angle bisectors of the angle *AOB* are given by

$$\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$$
$$\Rightarrow x^2 - y^2 = 0$$
$$\Rightarrow x + y = 0 \quad \text{and} \quad x - y = 0$$

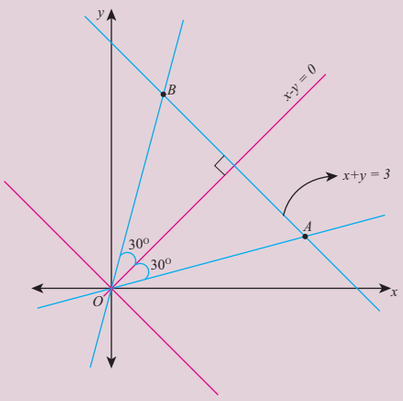


Figure 6.50

The angle bisectors $x - y = 0$ is perpendicular to the given line through *AB*,

that is $x + y = 3 \Rightarrow \Delta OAB$ is isosceles.
$\Rightarrow \angle ABO = \angle BAO = 60^\circ$

therefore the given lines form an equilateral triangle.

Example 6.37 If the pair of lines represented by $x^2 - 2cxy - y^2 = 0$ and $x^2 - 2dxy - y^2 = 0$ be such that each pair bisects the angle between the other pair, prove that $cd = -1$.

Solution:

Given that the pair of straight lines,

$$x^2 - 2cxy - y^2 = 0 \tag{6.37}$$
$$x^2 - 2dxy - y^2 = 0 \tag{6.38}$$